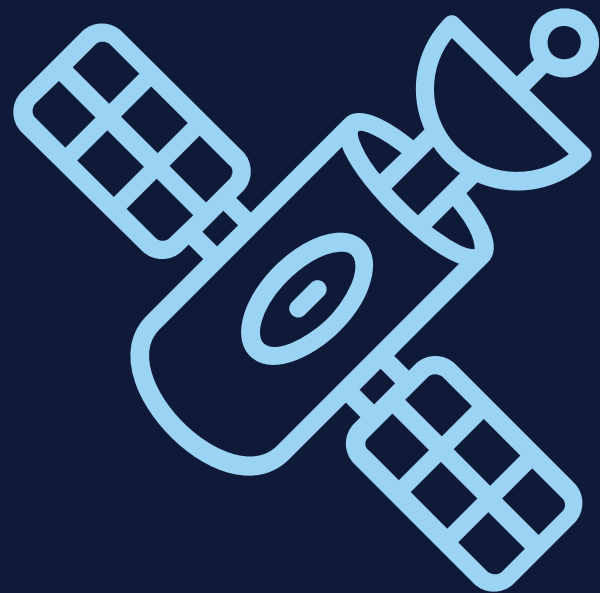


From Launch to Litter: Orbital Overload



Table Of Contents



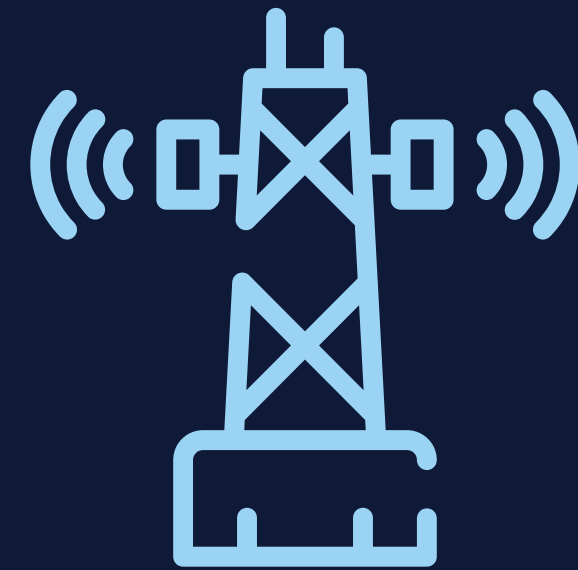
What is the Kessler Syndrome?

And its potential impact on civilization.



Exploring the Data

Looking at what's in orbit and who put it there.



Planning for The Future

How can we avoid a disaster?

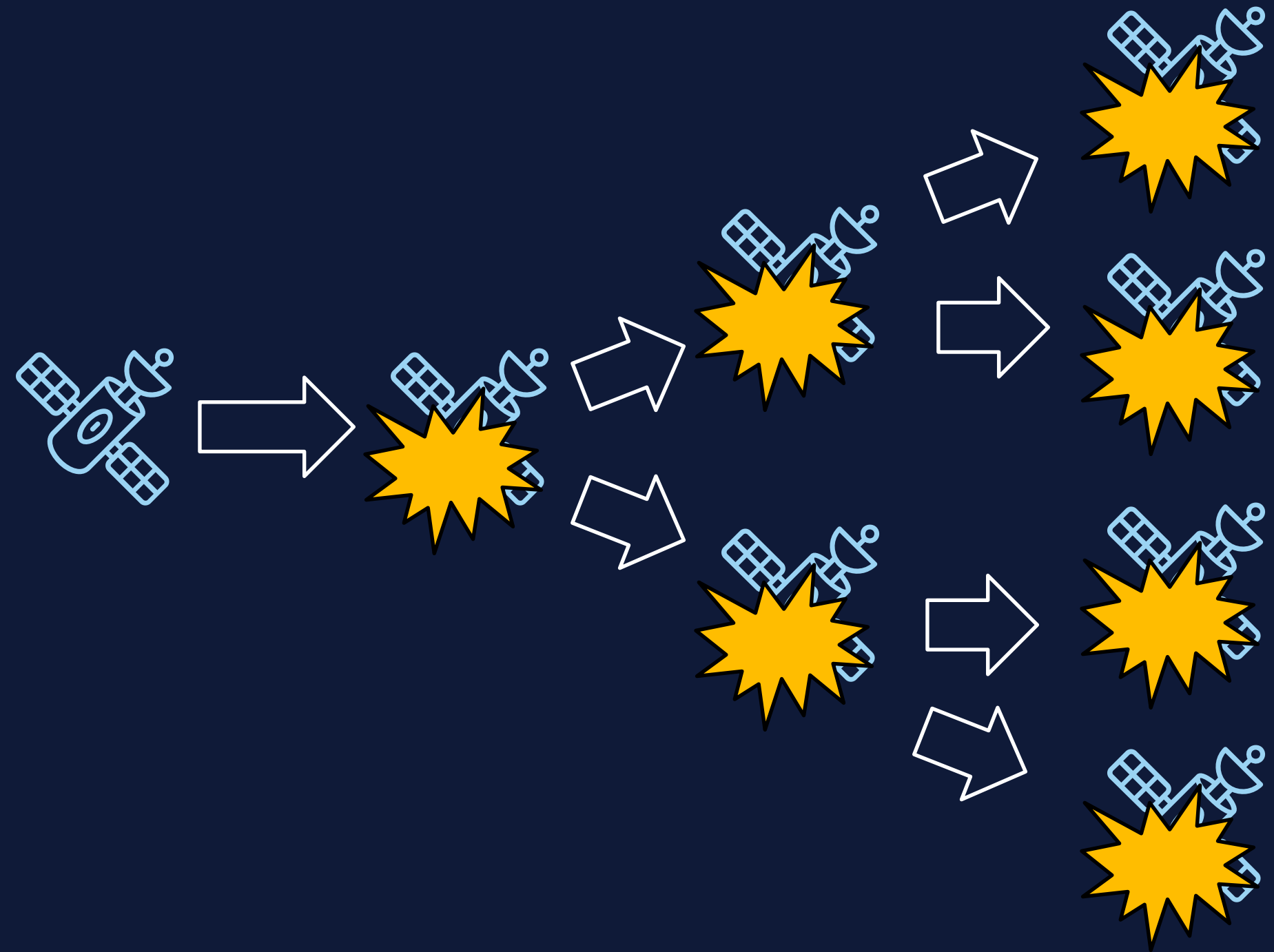
What is the **Kessler Syndrome**?

Collision Cascade

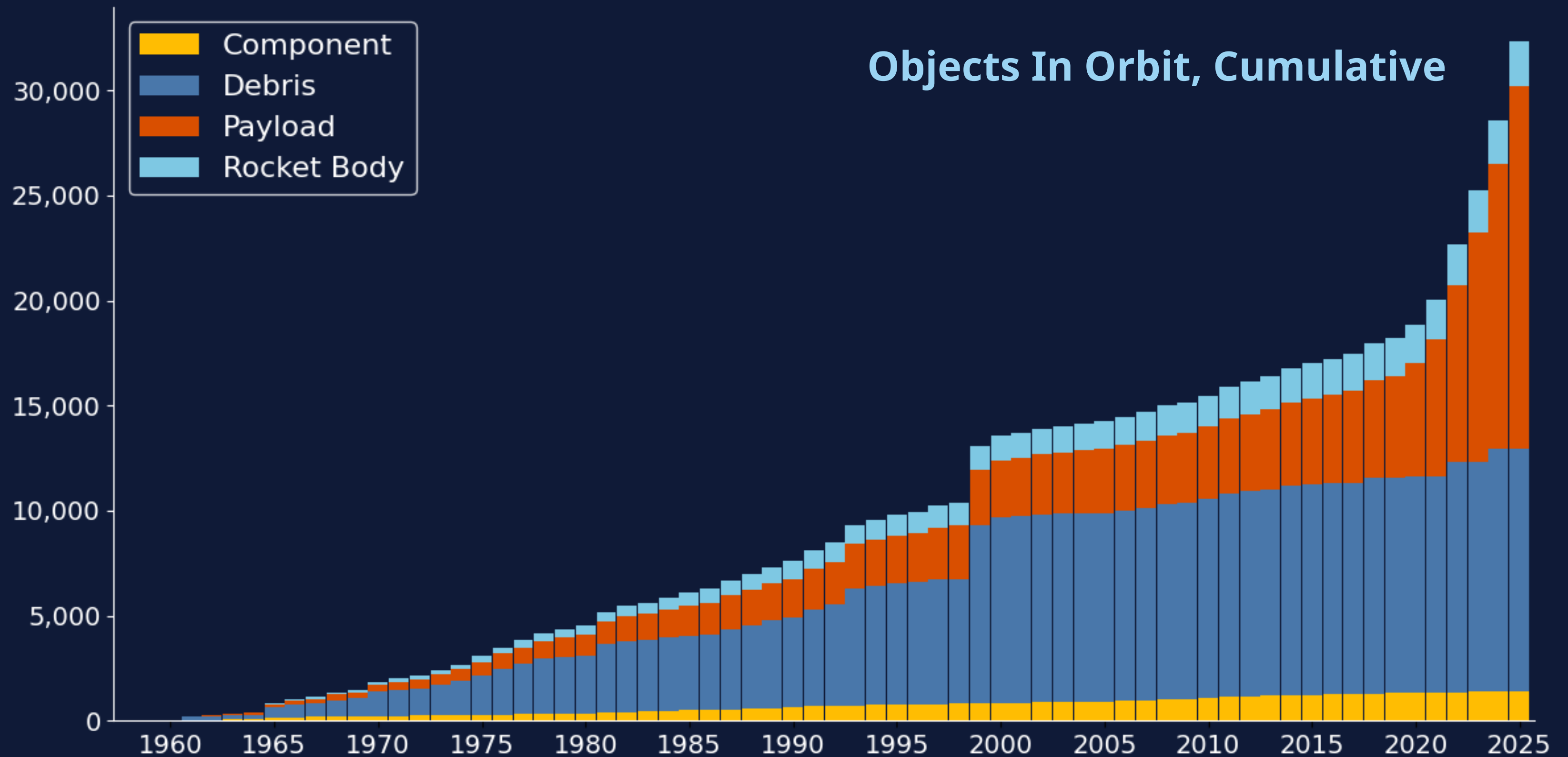
A scenario where satellite and/or space debris collisions cause a cascade effect, making low Earth orbit unusable.

Disaster Scenario

First identified in 1978 by Donald J. Kessler and Burton G. Cour-Palais.

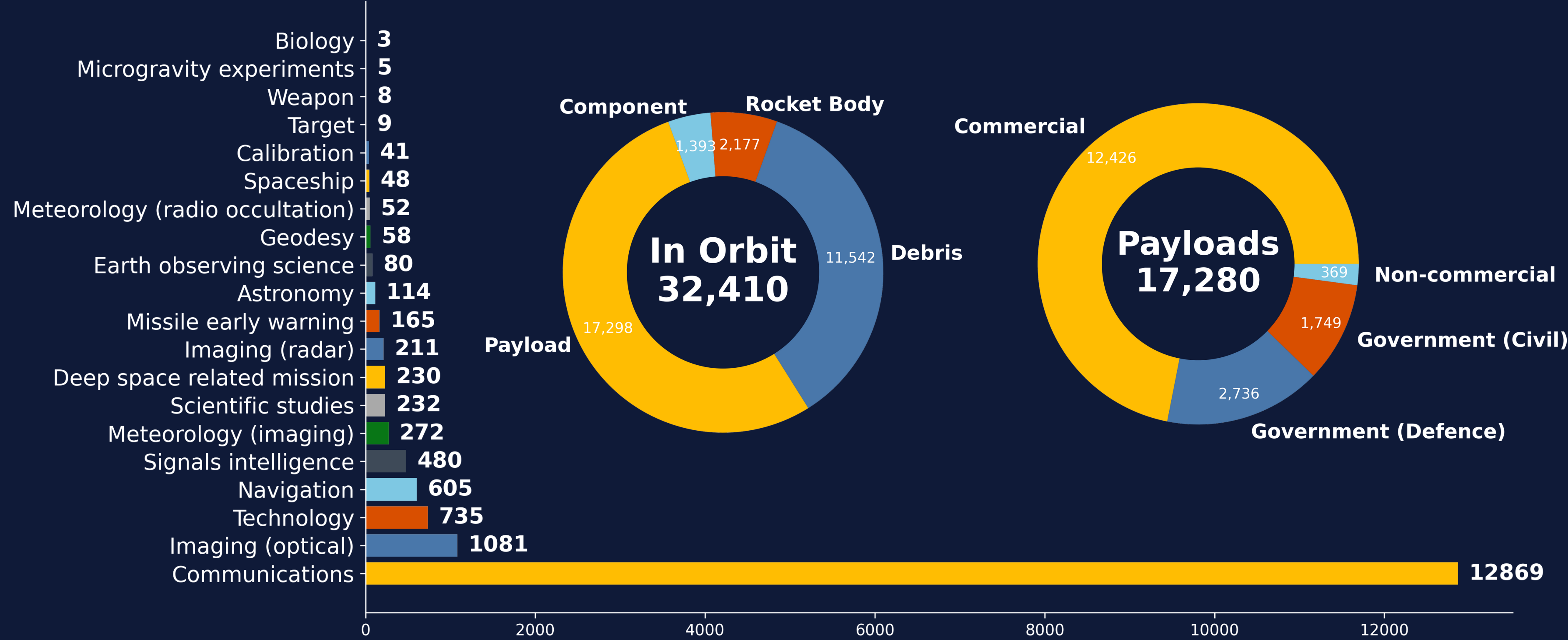


Satellite Launch Trends

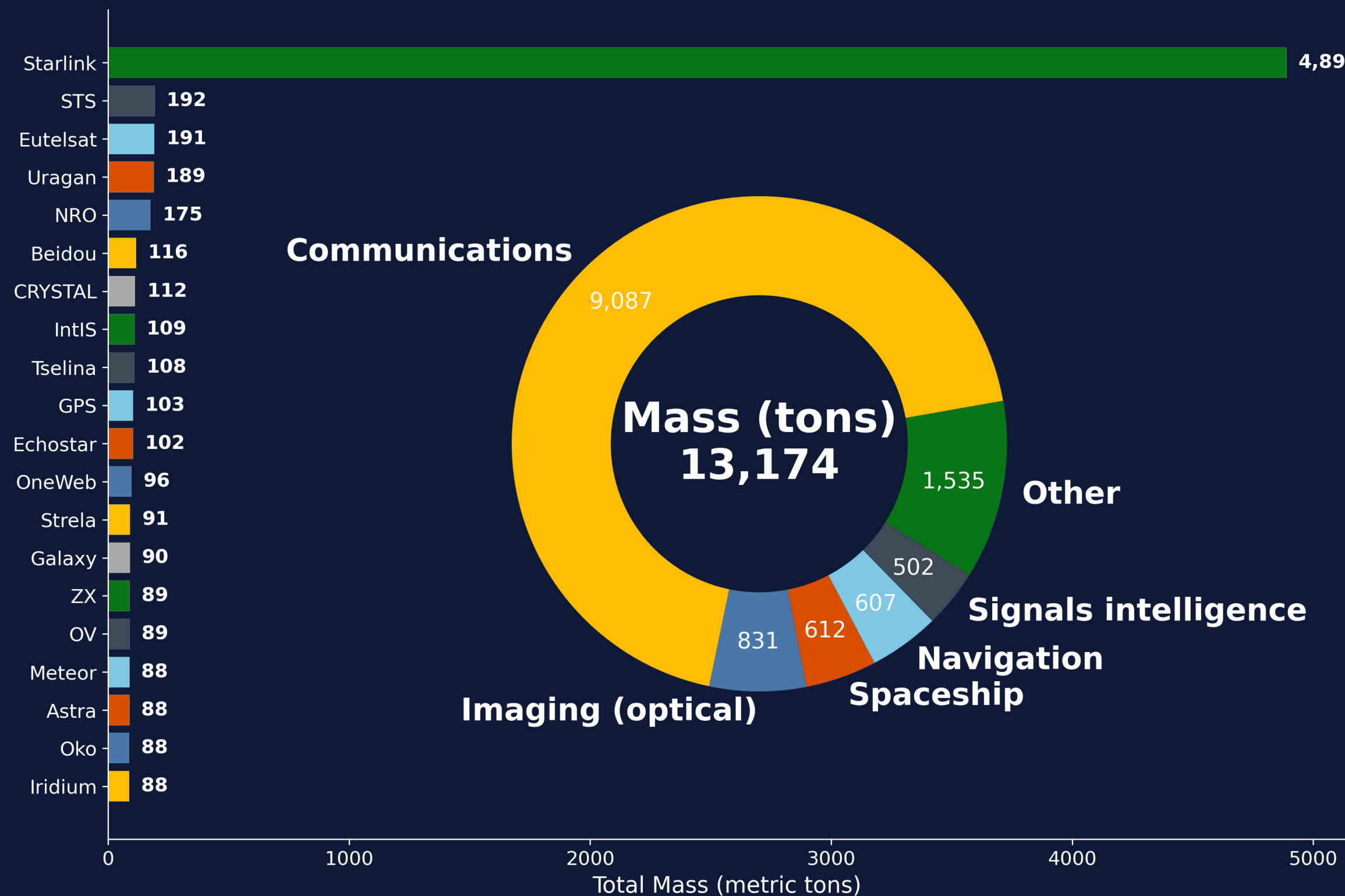


What's up in Orbit?

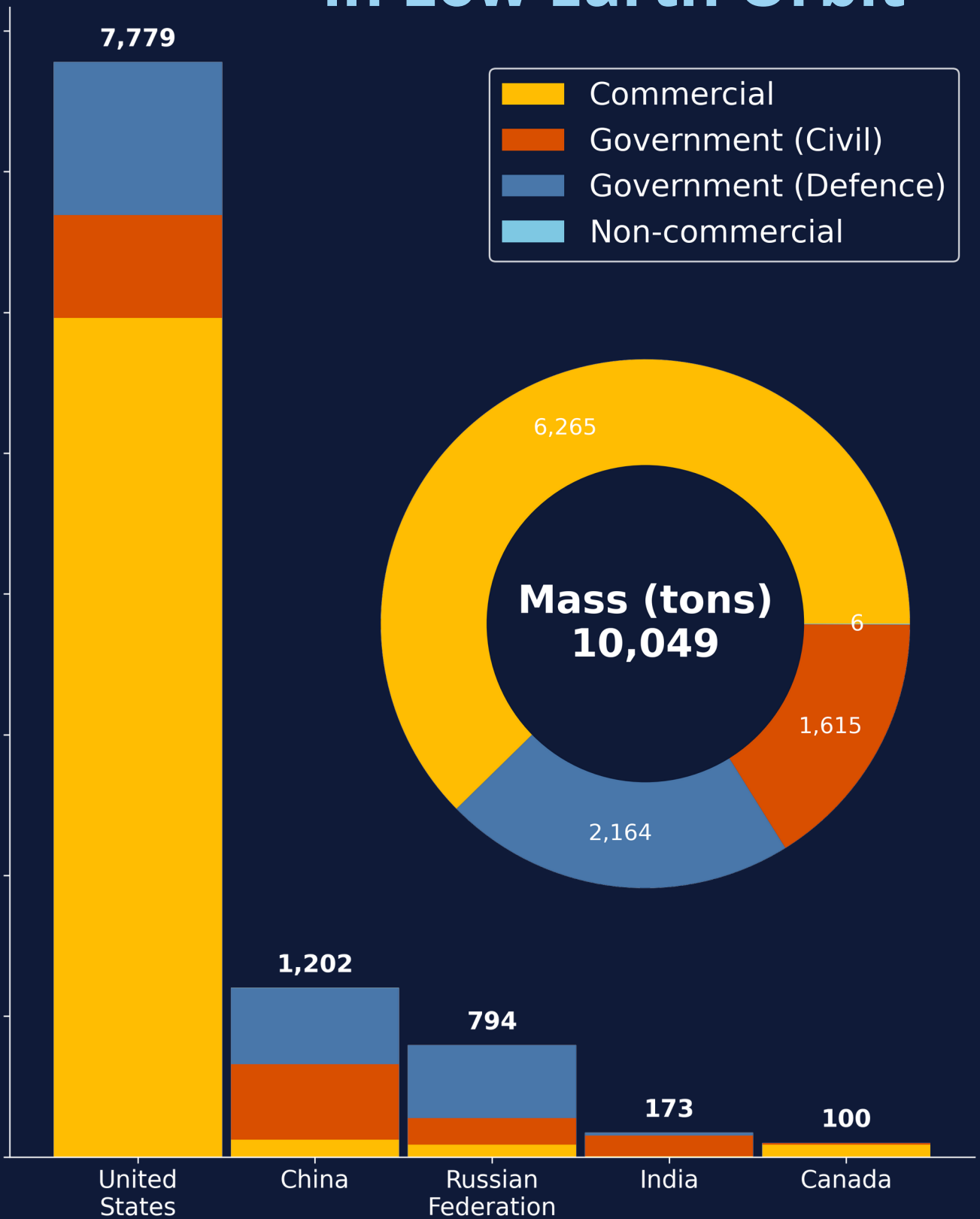
Count of Objects In Low Earth Orbit



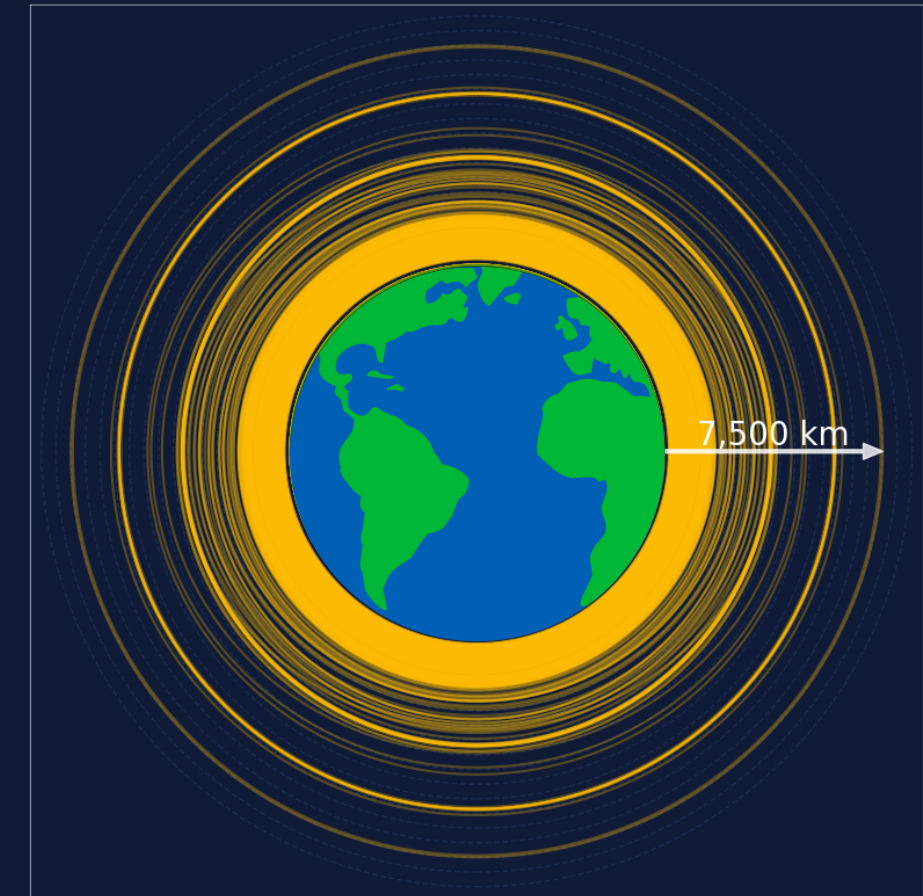
Heavy Metal



Mass of Payloads in Low Earth Orbit



Where are **they** all?



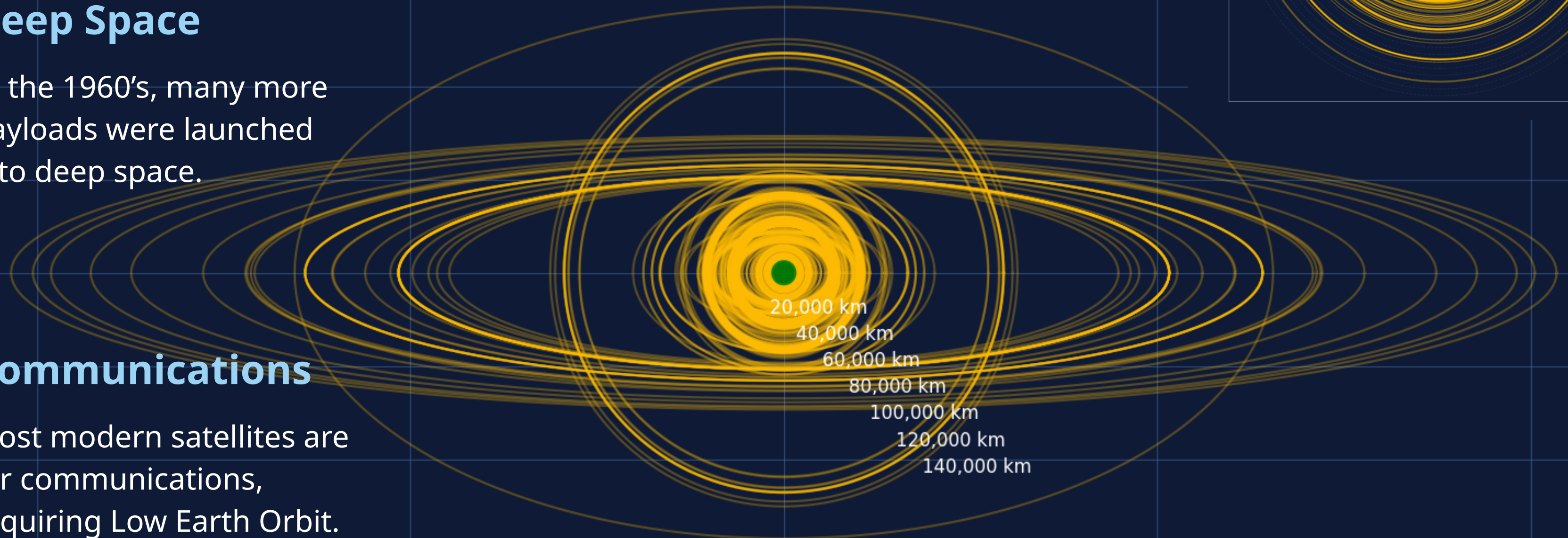
Orbital Paths of Payloads currently in Orbit

Deep Space

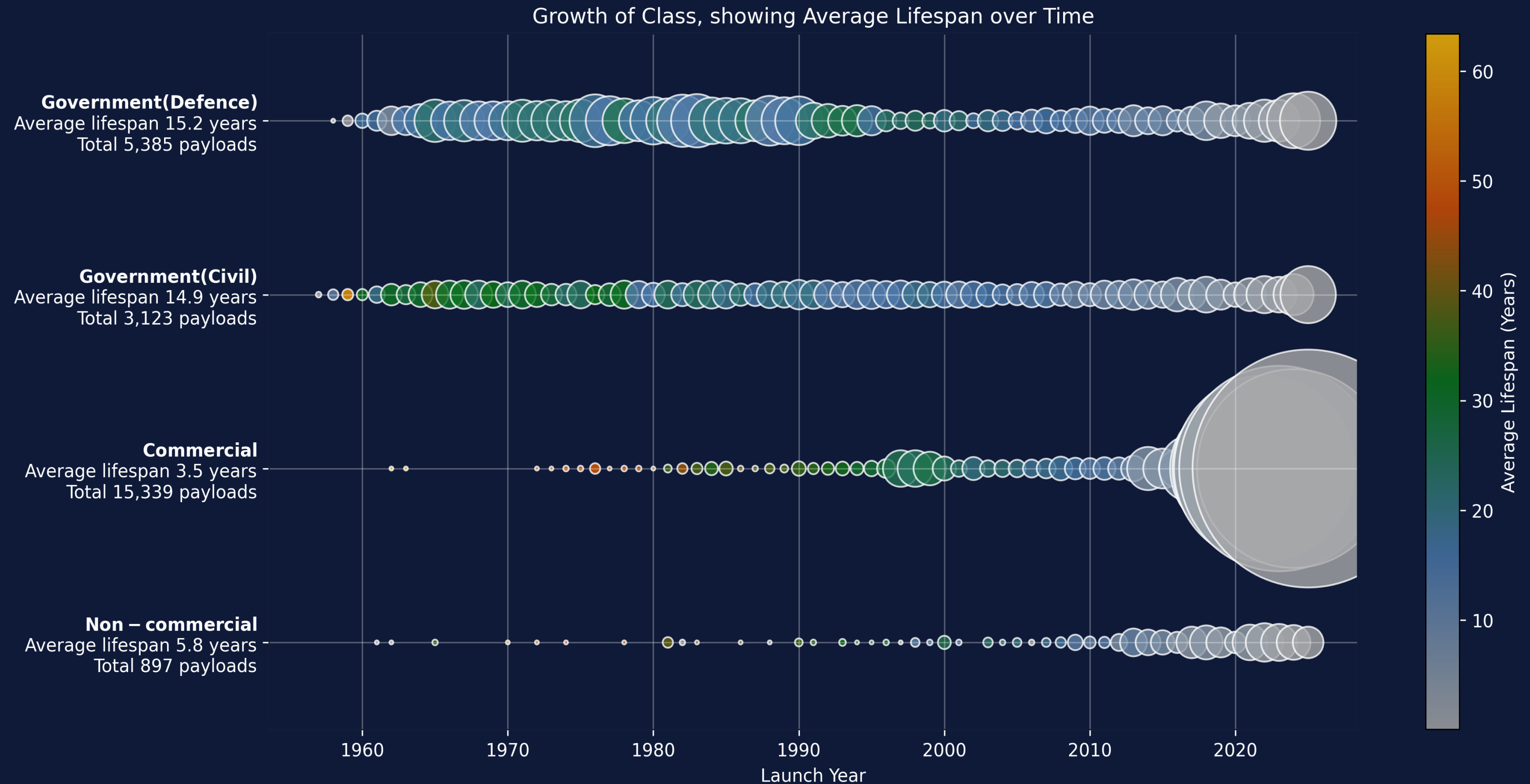
In the 1960's, many more payloads were launched into deep space.

Communications

Most modern satellites are for communications, requiring Low Earth Orbit.



Follow the Money



Operators and Sponsors

Starlink

First launch: **2018**

Total launches: **10,413**

Purpose: **Comms**

OneWeb

First launch: **2019**

Total launches: **656**

Purpose: **Internet**

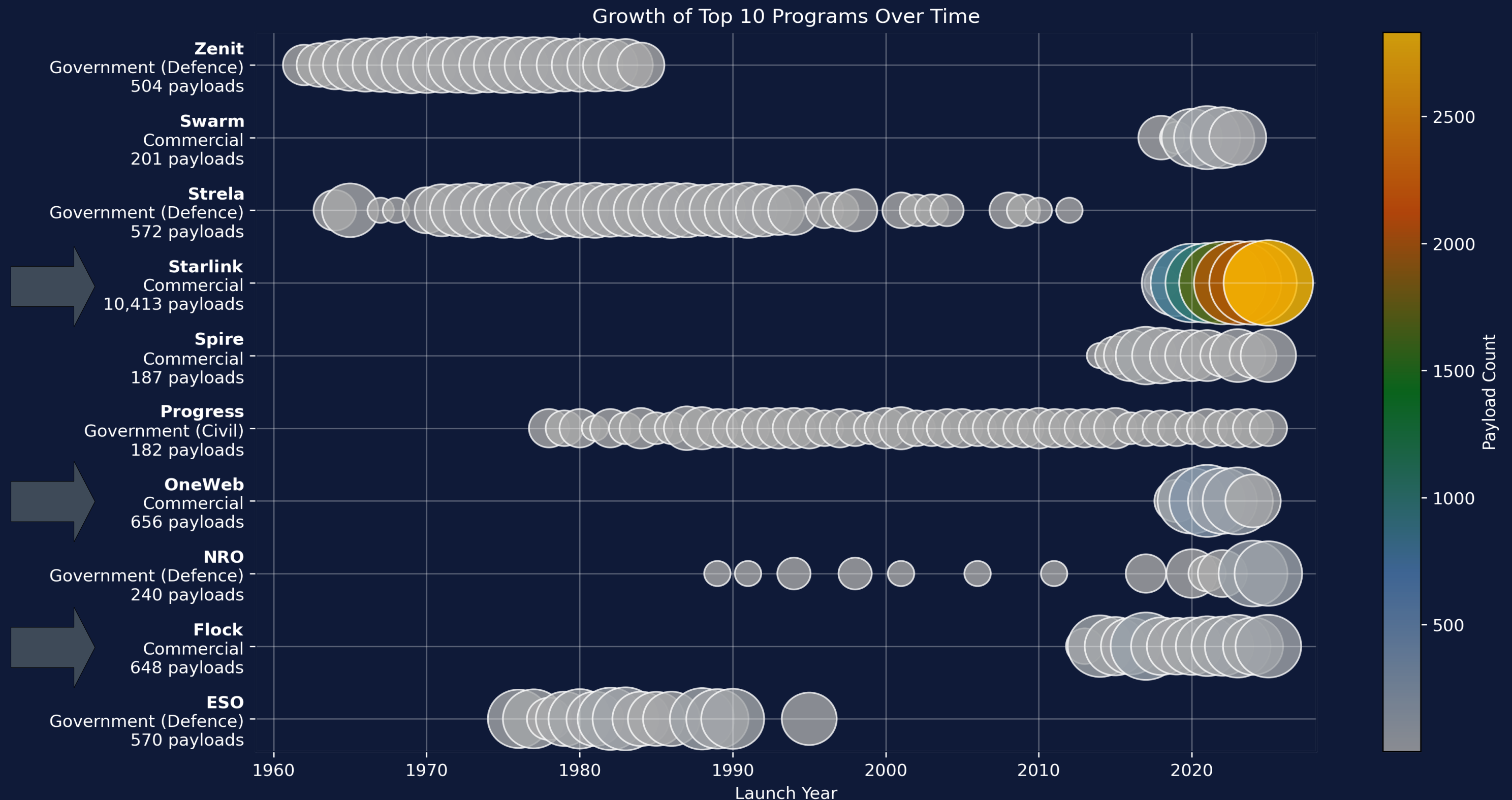
Flock

First launch: **2015**

Total launches: **548**

Purpose: **Earth**

Observation



Global Growth

1960

Total Countries: **2**

Total Satellites: 46

1980

Total Countries: **17**

Total Satellites: 2,834

2000

Total Countries: **39**

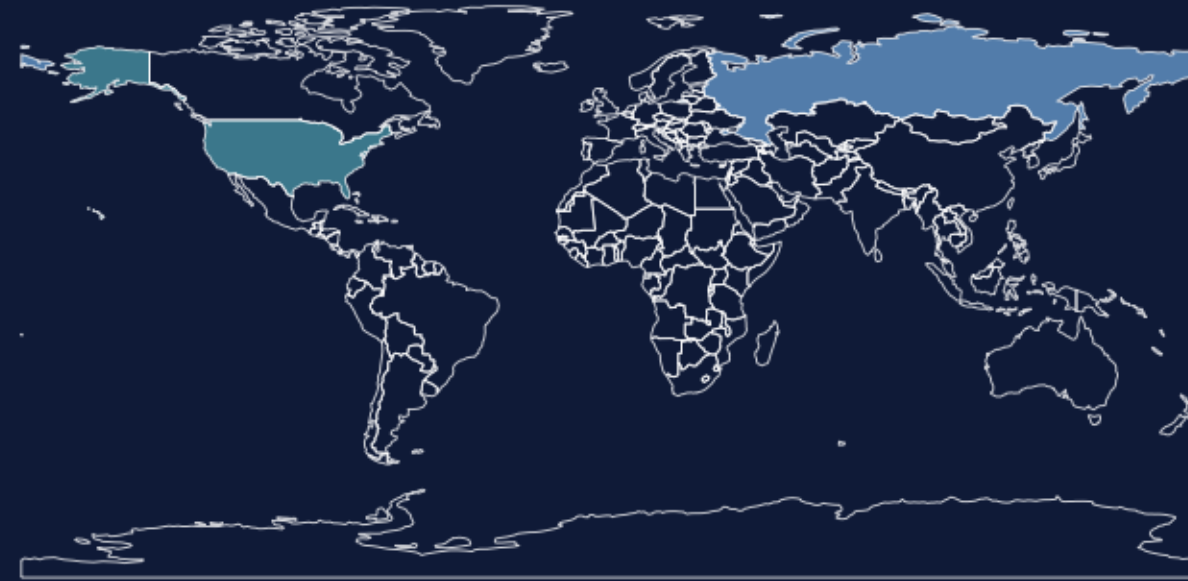
Total Satellites: 6,026

2025

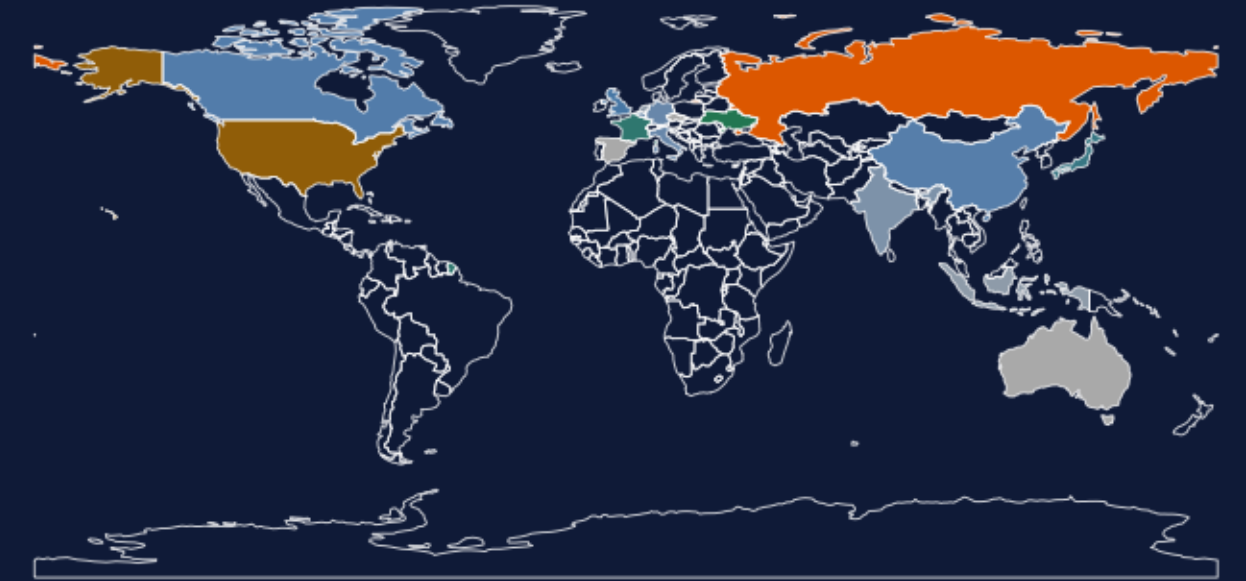
Total Countries: **98**

Total Satellites: 24,756

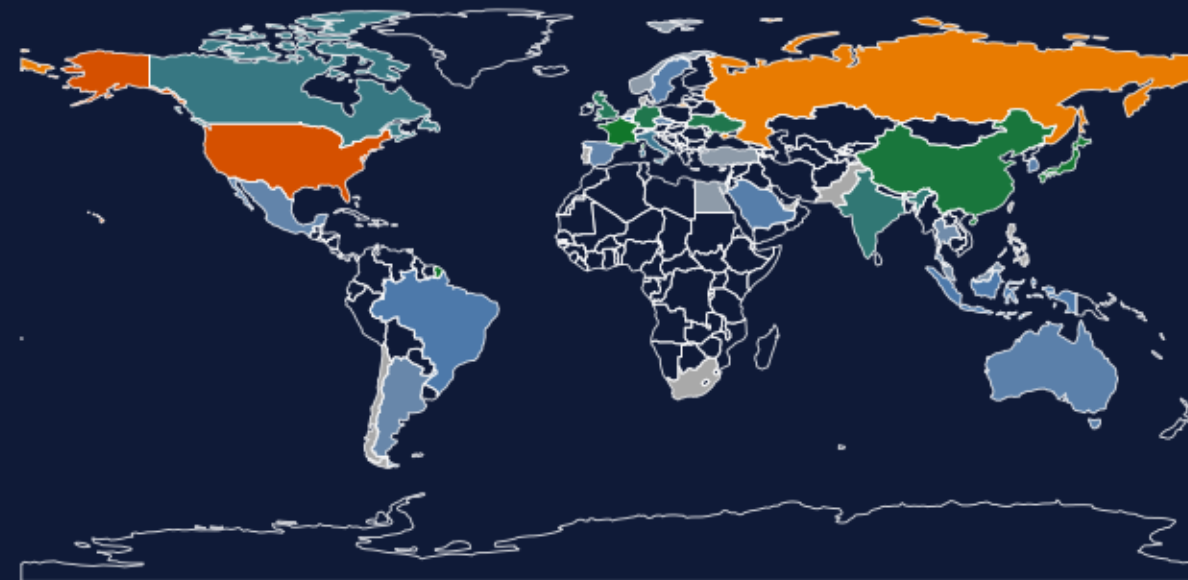
1960



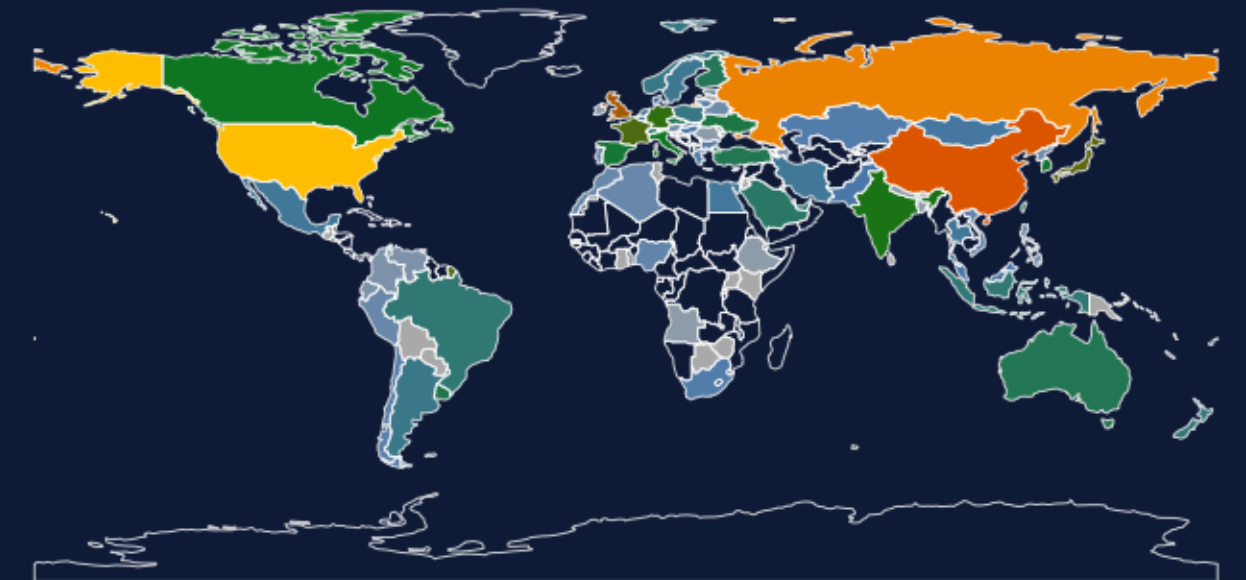
1980



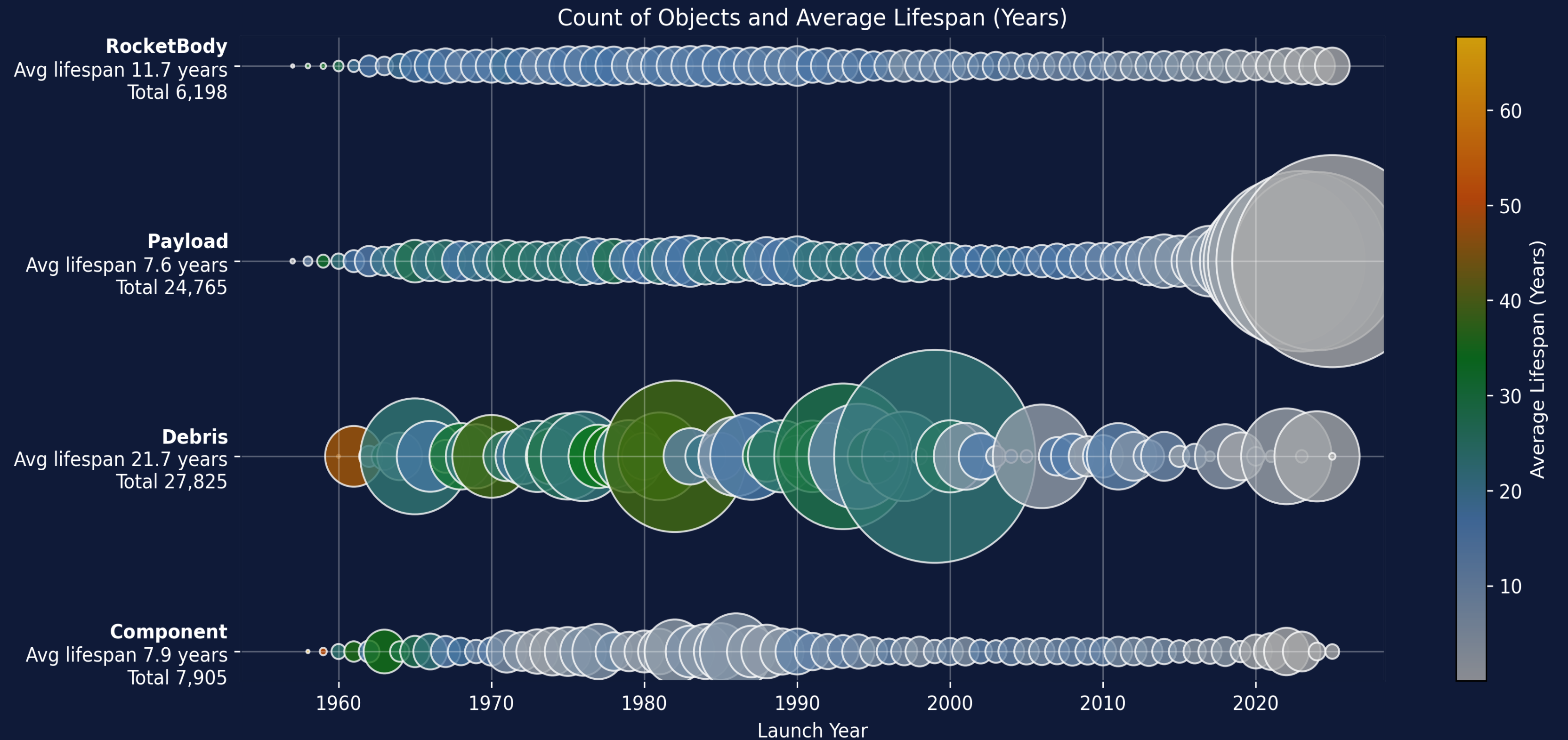
2000



2025



Long-lived Litter



What is **at Stake?**



A New Dark Age

Global Communications, GPS, Internet, TV,
Climate Change Monitoring and Emergency
Response – **All Gone**

Economic Disaster

Space Industry is worth \$400B.
Knock on effects to all Earth industry.

No More Missions

Space exploration would halt for safety reasons.
ISS, Tiangong abandoned.
Moon and Mars missions canceled.

Security and Defence

Loss of surveillance, weapons detection and
defence capability.

Stop it **and Tidy up**

Active Debris Removal

Capture and de-orbit defunct payloads.
Expensive.

Design for De-orbit

Satellites designed to decay and burn up.
Cheaper than recovery, but new payloads only.

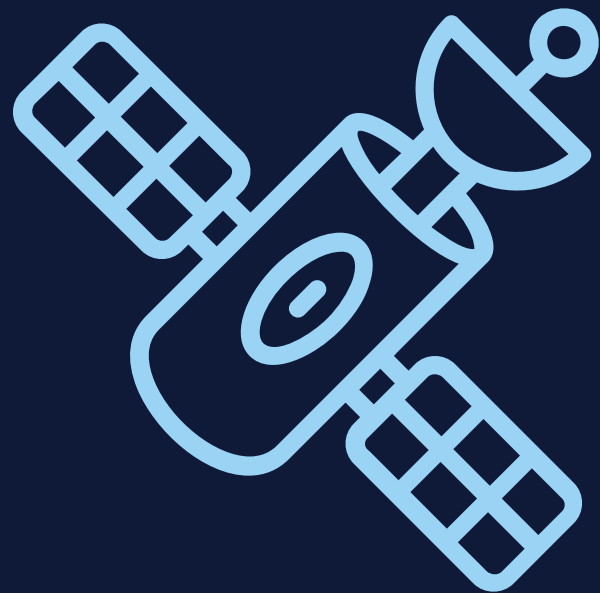
Collision Avoidance

Automated systems to monitor orbits and
adjust trajectory.

Policy and Agreements

International Guidelines (UN COPUOS).
Voluntary and weakly enforced.

In Conclusion



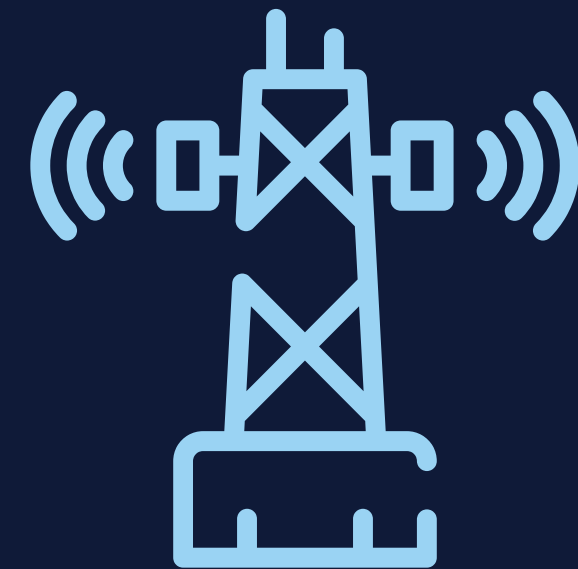
Devastating Kessler Cascade

A new dark age:
loss of navigation,
communications and
more.



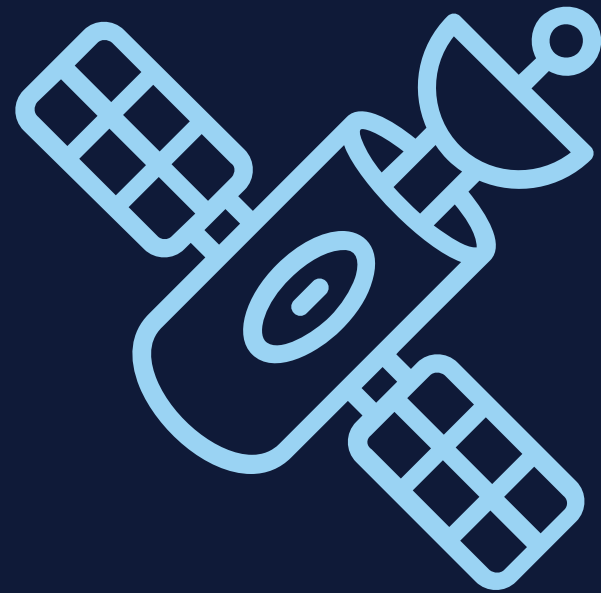
Low Earth Orbits are Crowded

There is a lot of hardware
in space, and we are
adding to it at
a record pace.



Planning for The Future

Without coordinated
efforts, we will reach a
tipping point within
decades, maybe years.



More Information:
<https://orbital.daemonchild.com>